Azure Local: the lifecycle, from discovery to Day 2

Handout for: Hybrid Cloud: Plan, Deploy, and Operate Azure Local from Discovery to Day 2 (NIC 2026).

This is the order the session followed, and the order an engineer can follow: thirteen stages, start to finish, on one two-node cluster. For each stage: what you do for Azure Local, where it is documented, which tools fit, and what the stage hands to the next one. The example values are the fictional company IIC, where you are the engineer, on two validated integrated-system nodes, switched storage, cloud witness, Local Identity with Key Vault, Azure Local 12.2609 (the latest at the time of the session; size against the release you deploy).

The three boundaries worth getting right

- **Prepare (7) is the environment, not the machines.** Landing zone, Key Vault or AD, switches, firewall, cabling. Deploy starts when the deployment gate is green.
- **Deploy (8) has three steps and ends when a VM can be created.** 8a provision the machines (firmware, OS, OS configuration, Arc registration). 8b deploy the cluster (Validate, then Deploy: cluster, storage pool, Network ATC intents, security baseline, Arc Resource Bridge). 8c post-deployment: the workload platform (workload volumes, logical networks and NSGs, storage paths, images).
- **Production goes live at a gate, not a stage.** Validate (9) checks what was built; Day-2 ready (10) makes it ownable; when the readiness gate is green, migrated workloads land (11) and new workloads are born under Operate (12).

The stages

#	Stage	What you do for Azure Local	Documentation	Tools	Hands on
1	Engage	Agree scope and outcomes; connected or disconnected; hyperconverged or disaggregated; AD-joined or Local Identity with Key Vault; name the owners (network, AD/DNS, firewall, OEM, operations).	Learn: *find your deployment type*. azurelocal.cloud: Planning & Discovery, Pre-Discovery Planning; Implementation 01 Purpose & Scope.	A repo as the single source	An agreed scope and the owners

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2	Assess	Inventory and classify the workloads: an RVTools export, or an Azure Migrate appliance (configuration, performance history, applications). Per workload: Azure Local VM, AVD, AKS, Arc service, Azure, later wave, or retire. RPO/RTO and licensing per tier. Name what the inventory cannot tell you: on the VMware to Azure Local route, agentless dependency analysis is not available, so dependencies come from the application owners and are recorded as assumptions.	Learn: *build a migration plan with Azure Migrate*. azurelocal.cloud: Requirements Workshops, Workload Assessment.	RVTools, Azure Migrate, Azure Local Surveyor	An inventory with its gaps named
3	Discover	The site and the enterprise around it: rack, power, cooling, the ToR switch pair and its RDMA capability, uplinks; the port map proven from the switch MAC and LLDP tables, never from labels; AD/DNS/NTP, proxy or Arc gateway, who approves firewall changes. Run the Environment Checker standalone before hardware arrives and hand the same endpoint data file to the firewall administrator. Decide the Arc gateway now: it cannot be enabled after deployment.	Learn: *physical network requirements*, *firewall requirements*, *Environment Checker*. azurelocal.cloud: Discovery Checklist, Site Assessment.	Environment Checker, Beacon (from the node, later), Ranger (brownfield only)	A site assessment report and a proven port map
4	Plan	Size with a node away (N+1): everything must fit one node with its reserve on two nodes. Hardware class from the Azure Local Catalog (Premier Solution, Integrated System, Validated Node). Node count and topology (switchless up to four nodes, switched, Rack Aware, disaggregated SAN). Network reference pattern and Network ATC intents from the proven port map. IP plan. Identity. Connectivity (direct, proxy, Arc gateway). Landing zone and naming. Continuity: backup for every tier, Site Recovery where minutes matter. Operating requirements: monitoring cost ceiling, maintenance windows, who updates what.	Learn: *network reference patterns*, *two-node IP requirements*, *baseline reference architecture: instance design choices*, the Catalog and Sizer. azurelocal.cloud: Hardware Requirements, Multi-Site Planning, Landing Zone Strategy, Naming Standards.	Azure Local Surveyor, Azure Local Catalog / Sizer, Network ATC intent model, Trailwright	The decisions, recorded; the bill of materials

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5	Design	HLD and LLD: cabling and port-to-intent map, switch configuration (VLANs, MTU, PFC/ETS for RoCE), storage layout and the repair reserve (pool capacity equal to one capacity drive per node, kept free), logical networks and NSGs, OU and naming, RBAC, Key Vault, Log Analytics workspace, Recovery Services vault, witness.	Learn: *host network requirements*, *system requirements*, *baseline reference architecture*. azurelocal.cloud: High-Level Design, Low-Level Design.	Trailwright, your diagramming tool	The HLD and LLD
6	Document and hand off	The design record validated; `infrastructure.yml` or the parameter file exported; Microsoft's deployment checklist filled in (machine names, OU DN or none, domain, DNS servers, credentials, IP ranges, VLANs, custom location, witness); the firewall allow-list as a data file; go/no-go.	Learn: *deployment prerequisites and checklist*. azurelocal.cloud: Deliverables Handoff, 03 Key Inputs & Variables.	Trailwright, Bicep what-if, the repo	The design record, `infrastructure.yml`, the checklist
7	Prepare the environment	Azure: landing zone (resource groups, providers, RBAC, Log Analytics, Key Vault, witness storage, Arc gateway resource), Policy assigned, PIM roles eligible. Identity: a Key Vault for Local Identity, or the AD preparation module (OU, deployment user, GPO block). Site: racked and cabled per the port map, switch configuration, firewall rules in place, firmware at the OEM baseline. The deployment gate: every line true before any node is touched.	Learn: *prepare Active Directory*, *Arc registration permissions*. azurelocal.cloud: Implementation Parts 2 and 3.	Bicep (landing zone), Key Vault, Environment Checker, Azure Policy, Entra PIM	A landing zone, a vault, a proven site
8a	Deploy: provision the machines	Firmware and drivers at the Solution Builder Extension baseline; OS install per node (ISO, or simplified machine provisioning, preview); OS configuration (name, static IP, DNS, NTP, local administrator, SSH); Arc registration (direct, proxy or gateway) and the mandatory extensions; both machines Ready. An older image runs the OS update during Arc registration (40 to 45 minutes).	Learn: *install the OS*, *register with Arc*. azurelocal.cloud: Part 4 phases 01–04.	Beacon, ZTP, Environment Checker, the OEM SBE, Azure Arc	Arc-registered machines

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8b	Deploy: the cluster	Portal wizard or ARM/Bicep, the same deployment package. Validate (about 15 minutes), then Deploy (about 2.5 hours for two nodes): the failover cluster, the Storage Spaces Direct pool and infrastructure volumes, the Network ATC intents, the security baseline with drift control, the Arc Resource Bridge (40 to 45 minutes on its own), the custom location, the witness. Local Identity: nodes stay in WORKGROUP, secrets in the vault.	Learn: *deploy via portal*, *deploy via ARM template*, *Rack Aware deployment*. azurelocal.cloud: Part 4 phase 05.	Azure portal or Bicep, Bicep what-if, Sonar, Network ATC	A deployed, Arc-managed cluster
8c	Deploy: the workload platform	Workload volumes and storage paths; logical networks with their VLANs and NSGs (SDN enabled by Arc is permanent: a design decision); VM images; health alerts on; RDP stays off. Ends when a VM can be created.	Learn: *post-deployment tasks*, *logical networks*, *storage paths*, *VM images*. azurelocal.cloud: Part 4 phase 06.	Azure Arc VM platform, Bicep	A platform a VM can be created on
9	Validate	Check, do not build: the resource and both machines Connected, Resource Bridge and custom location, `Get-NetIntentStatus` Success, storage health, the security baseline compliance (99 % expected on Secured-core hardware), a first VM through Arc, benchmarks if sizing must be proven, the as-built recorded against the design. Run the operational readiness gate once: expect red.	Learn: *verify a successful deployment*, *manage security defaults*. azurelocal.cloud: Part 6 Testing & Validation, Load Testing.	Post-deployment checks, Network ATC, Sonar replay, Ranger (as-built mode)	An as-built record, the Day-2 to-do list

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10	Day-2 ready	Monitoring: Insights with a DCR sized to the cost ceiling, alert rules and an action group, VM Insights on the workload VMs. Updates: Azure Update Manager maintenance configuration; Lifecycle Manager for the cluster, Update Manager for guests, Intune for Windows 11. Protection: backup (Azure Backup or a partner product such as Veeam, Commvault, Cohesity or Rubrik for Azure Local VMs) and Site Recovery with a test failover. Access: RBAC with Entra PIM, eligible not standing. Governance: Defender for Cloud and the Azure Policy baseline. The readiness gate: every control Verified with evidence, then the hand-off to the ops team (runbook, as-built package, alert routes, maintenance window).	Learn: *Insights*, *health alerts*, *Azure Update Manager*, *Defender for Cloud*, *backup options for Azure Local VMs*, *Site Recovery on Azure Local*. azurelocal.cloud: Part 5 Operational Foundations.	Azure Monitor and DCRs, Azure Update Manager, Azure Backup / Site Recovery, Entra PIM, Azure Policy, Defender for Cloud, Ranger (as-built package)	Verified controls, the runbook, the go-live
—	Gate: production go-live	The readiness gate is green and the ops team owns the platform. Nothing production lands before this line.			
11	Migrate	Azure Migrate to Azure Local in waves from the Assess complexity scores: target appliance on the cluster, source appliance on vCenter or Hyper-V, replicate to a logical network and storage path, migrate with the source shut down, verify, Complete migration, turn guest management on so the landed VMs join monitoring, updates, backup and policy. Partner tools where VDDK or the Hyper-V preview blocks.	Learn: *migration options*, *VMware* and *Hyper-V with Azure Migrate*. azurelocal.cloud: VMware Migration, Nutanix Migration, VM Conversion Toolkit.	Azure Migrate, Surveyor (waves)	Wave 1 running under the Day-2 scope

#	Stage	What you do for Azure Local	Documentation	Tools	Hands on
12	Operate	Solution updates through Update Manager (readiness checks, SBE in the same run, one node drained at a time; stay within six months of the current release); daily health checks routed to the ops process; faults handled as symptom, signal, diagnosis, action, recovery evidence (drive repair, node repair, Network ATC drift); capacity as a decision (on two nodes, full means the node-loss reserve is gone); the VM lifecycle through Arc; DR drills and restore tests; a current-state snapshot before and after every change.	Learn: *update phases*, *maintain servers*, *repair a node*, *add a node*, *VM operations*. azurelocal.cloud: Operations.	Lifecycle Manager via Update Manager, Azure Monitor, Observability Agent (preview), Network ATC, Azure Arc, Site Recovery, Ranger (current state)	A cluster that stays up, and evidence for Review
13	Review and optimize	Document the whole solution as it is now (Ranger current-state, diffed against the as-built); right-size from measured utilisation; compliance and update history; the DR test record; DCR cost against the ceiling; a Well-Architected review; the capacity decision fed back to Plan.	Learn: *architecture best practices for Azure Local*, Azure Local LENS. azurelocal.cloud: Governance and Compliance, Cost Management, Load Testing.	Ranger, Surveyor, Insights, Policy reports	The next plan

Where to find the documentation

- Microsoft Learn, Azure Local: https://learn.microsoft.com/azure/azure-local/
- azurelocal.cloud documentation (Planning & Discovery, Design, Implementation Guide, Operations): https://azurelocal.cloud/docs/
- Azure Local Surveyor: https://azurelocal.cloud/azurelocal-surveyor/ · Trailwright: https://labs.hybridsolutions.cloud/azurelocal-trailwright/ · Beacon: https://azurelocal.cloud/azurelocal-beacon/ · Sonar: https://azurelocal.cloud/azurelocal-sonar/ · Ranger: https://azurelocal.cloud/azurelocal-ranger/
- The session repo: https://github.com/thisismydemo/nic2026_azl